

Configuring a Basic Spring Application

Scenario

Your company is developing a web application for managing a library. You need to use the Spring Framework to handle the backend operations.

Step 1: Create a Maven Project

Project Name:

LibraryManagement

Step 2: Add Spring Core Dependency (**pom.xml**)

```
<dependencies>
```

```
    <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-context</artifactId>
        <version>5.3.30</version>
    </dependency>
```

```
</dependencies>
```

Step 3: Create **applicationContext.xml**

Place inside:

src/main/resources/applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="
        http://www.springframework.org/schema/beans
        http://www.springframework.org/schema/beans/spring-beans.xsd">
```

```
    <bean id="bookRepository"
        class="com.library.repository.BookRepository"/>
```

```
<bean id="bookService"
      class="com.library.service.BookService"/>

</beans>
```

Step 4: BookRepository.java

```
package com.library.repository;

public class BookRepository {

    public void displayBooks() {
        System.out.println("Displaying books from repository.");
    }
}
```

Step 5: BookService.java

```
package com.library.service;

public class BookService {

    public void showService() {
        System.out.println("Book Service is running.");
    }
}
```

Step 6: Main Class

```
package com.library;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

import com.library.service.BookService;

public class LibraryManagementApplication {

    public static void main(String[] args) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");
```

```
    BookService service =  
        (BookService) context.getBean("bookService");  
  
    service.showService();  
}  
}
```

Output

Book Service is running.

Implementing Dependency Injection

Step 1: Update **applicationContext.xml**

```
<?xml version="1.0" encoding="UTF-8"?>  
  
<beans xmlns="http://www.springframework.org/schema/beans"  
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
    xsi:schemaLocation="  
        http://www.springframework.org/schema/beans  
        http://www.springframework.org/schema/beans/spring-beans.xsd">  
  
    <bean id="bookRepository"  
        class="com.library.repository.BookRepository"/>  
  
    <bean id="bookService"  
        class="com.library.service.BookService">  
  
        <property name="bookRepository"  
            ref="bookRepository"/>  
  
    </bean>  
  
</beans>
```

Update BookService.java

```
package com.library.service;  
  
import com.library.repository.BookRepository;
```

```
public class BookService {  
  
    private BookRepository bookRepository;  
  
    public void setBookRepository(BookRepository bookRepository) {  
        this.bookRepository = bookRepository;  
    }  
  
    public void showService() {  
  
        System.out.println("Book Service Started.");  
  
        bookRepository.displayBooks();  
    }  
}
```

BookRepository.java

```
package com.library.repository;  
  
public class BookRepository {  
  
    public void displayBooks() {  
  
        System.out.println("Fetching books from database.");  
    }  
}
```

Output

Book Service Started.
Fetching books from database.

Implementing Logging with Spring AOP

Step 1: Add Dependencies

<dependencies>

<dependency>

<groupId>org.springframework</groupId>

<artifactId>spring-context</artifactId>

```

        <version>5.3.30</version>
    </dependency>

    <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-aop</artifactId>
        <version>5.3.30</version>
    </dependency>

    <dependency>
        <groupId>org.aspectj</groupId>
        <artifactId>aspectjweaver</artifactId>
        <version>1.9.20</version>
    </dependency>

</dependencies>

```

LoggingAspect.java

```

package com.library.aspect;

import org.aspectj.lang.ProceedingJoinPoint;

public class LoggingAspect {

    public Object logExecutionTime(ProceedingJoinPoint joinPoint)
        throws Throwable {

        long start = System.currentTimeMillis();

        Object result = joinPoint.proceed();

        long end = System.currentTimeMillis();

        System.out.println(joinPoint.getSignature().getName()
            + " executed in "
            + (end - start)
            + " ms");

        return result;
    }
}

```

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:aop="http://www.springframework.org/schema/aop"

xsi:schemaLocation="
http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd

http://www.springframework.org/schema/aop
http://www.springframework.org/schema/aop/spring-aop.xsd">

<bean id="bookRepository"
class="com.library.repository.BookRepository"/>

<bean id="bookService"
class="com.library.service.BookService">

<property name="bookRepository"
ref="bookRepository"/>

</bean>

<bean id="loggingAspect"
class="com.library.aspect.LoggingAspect"/>

<aop:config>

<aop:aspect ref="loggingAspect">

<aop:around
method="logExecutionTime"
pointcut="execution(* com.library.service.*(..))"/>

</aop:aspect>

</aop:config>

</beans>
```

Output

Book Service Started.
Fetching books from database.
showService executed in 5 ms

Creating and Configuring a Maven Project

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0">  
  
  <modelVersion>4.0.0</modelVersion>  
  
  <groupId>com.library</groupId>  
  
  <artifactId>LibraryManagement</artifactId>  
  
  <version>1.0</version>  
  
  <dependencies>  
  
    <dependency>  
  
      <groupId>org.springframework</groupId>  
  
      <artifactId>spring-context</artifactId>  
  
      <version>5.3.30</version>  
  
    </dependency>  
  
    <dependency>  
  
      <groupId>org.springframework</groupId>  
  
      <artifactId>spring-aop</artifactId>  
  
      <version>5.3.30</version>  
  
    </dependency>  
  
  </dependencies>  
</project>
```

```
<groupId>org.springframework</groupId>

<artifactId>spring-webmvc</artifactId>

<version>5.3.30</version>

</dependency>

</dependencies>

<build>

<plugins>

<plugin>

<groupId>org.apache.maven.plugins</groupId>

<artifactId>maven-compiler-plugin</artifactId>

<version>3.8.1</version>

<configuration>

<source>1.8</source>

<target>1.8</target>

</configuration>

</plugin>

</plugins>

</build>

</project>
```

Output

BUILD SUCCESS

Configuring the Spring IoC Container

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"

xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

xsi:schemaLocation="

http://www.springframework.org/schema/beans

http://www.springframework.org/schema/beans/spring-beans.xsd">

<bean id="bookRepository"

class="com.library.repository.BookRepository"/>

<bean id="bookService"

class="com.library.service.BookService">

<property name="bookRepository"

ref="bookRepository"/>

</bean>

</beans>
```

BookRepository.java

```
package com.library.repository;

public class BookRepository {

    public void displayBooks() {

        System.out.println("Book Repository Working...");

    }

}
```

BookService.java

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {

    private BookRepository bookRepository;

    public void setBookRepository(BookRepository bookRepository) {

        this.bookRepository = bookRepository;
    }

    public void showService() {

        System.out.println("Book Service Working...");

        bookRepository.displayBooks();
    }
}
```

LibraryManagementApplication.java

```
package com.library;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

import com.library.service.BookService;

public class LibraryManagementApplication {

    public static void main(String[] args) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");

        BookService service =
            context.getBean("bookService", BookService.class);

        service.showService();
    }
}
```

}

Output

Book Service Working...

Book Repository Working...